

Conservative Oxygen Therapy in Mechanically Ventilated Critically Ill Adult Patients

The UK-ROX Randomized Clinical Trial

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IMPORTANCE Supplemental oxygen is frequently given to patients in intensive care units (ICUs); however, there is insufficient evidence to guide its therapeutic use and to minimize the potential harm caused by administering too little or too much.

OBJECTIVE To determine whether reducing exposure to supplemental oxygen through a strategy of conservative oxygen therapy by using a peripheral oxygen saturation (Spo₂) target of 90% (range, 88%-92%) reduces mortality at 90 days in mechanically ventilated adult patients receiving supplemental oxygen in the ICU.

DESIGN, SETTING, AND PARTICIPANTS Multicenter, pragmatic, randomized clinical trial conducted in 97 ICUs in the UK including 16 500 mechanically ventilated patients receiving supplemental oxygen. Participants were enrolled between May 2021 and November 2024. Follow-up was completed in February 2025.

INTERVENTIONS Participants randomized to conservative oxygen therapy (n = 8258) received the lowest fraction of inspired oxygen possible to maintain their Spo₂ at 90%. Participants randomized to usual oxygen therapy (n = 8242) received oxygen therapy at the discretion of the treating clinician.

MAIN OUTCOMES AND MEASURES The primary outcome was all-cause mortality at 90 days. Secondary outcomes included duration of ICU and acute hospital stay among survivors, days alive and free from organ support at 30 days, and mortality at other time points.

RESULTS Of 16 500 randomized patients, primary outcome data were available for 16 394 (8211 in the conservative and 8183 in the usual oxygen therapy group). Randomized groups were similar (median age, 60 [IQR, 48-71] years and 38.2% females in both groups [n = 5652]). Exposure to supplemental oxygen was 29% lower for participants in the conservative oxygen therapy group compared with the usual oxygen therapy group. By 90 days, 2908 participants (35.4%) in the conservative oxygen therapy group had died compared with 2858 (34.9%) in the usual oxygen therapy group. After adjustment for prespecified baseline variables, the risk difference was 0.7 percentage points (95% CI, -0.7 to 2.0; P = .28). There were no significant differences in durations in ICU or hospital stay, days alive and free from organ support at 30 days, or mortality at other time points.

CONCLUSIONS AND RELEVANCE In adult patients receiving mechanical ventilation and supplemental oxygen in the ICU, minimizing oxygen exposure through conservative oxygen therapy did not significantly reduce all-cause mortality at 90 days.

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Oxygen is one of the most commonly administered treatments to patients in intensive care units (ICUs).¹ Traditionally, hypoxemia was avoided to minimize the risk of causing cellular hypoxia and organ dysfunction. In response to this, a liberal approach to oxygen therapy was commonplace for critically ill patients²; however, excessive administration of supplemental oxygen may also lead to harm.³ Achieving a balance between too little and too much oxygen could therefore be essential to optimize clinical outcomes for patients.

Clinical trials to date have been unable to determine whether administering less, rather than more, oxygen to patients is beneficial. Several clinical trials have shown no difference in outcome between a conservative compared with a more liberal approach to oxygen therapy in critically ill patients.⁴⁻⁸ However, in patients admitted to the ICU with COVID-19 and severe hypoxemia, conservative oxygen therapy resulted in more days alive without life support in the ICU.⁹ Moreover, in mechanically ventilated children receiving supplemental oxygen in a pediatric ICU, conservative oxygen therapy resulted in a reduction in a composite of organ support at 30 days or death.¹⁰ A recent systematic review and meta-analyses of 13 clinical trials including 10 632 adult patients reported no significant mortality difference between conservative and liberal oxygen therapy.¹¹ For such a widely used intervention, even a small survival benefit could translate into large numbers of lives saved. Therefore, further evidence from large-scale trials is required to determine whether conservative oxygen therapy is beneficial to patients receiving mechanical ventilation.

The UK Intensive Care Unit Randomized Trial Comparing Two Approaches to Oxygen Therapy (UK-ROX) trial assessed whether reducing exposure to supplemental oxygen through a strategy of conservative oxygen therapy by targeting a peripheral oxygen saturation (SpO₂) of 90% (range, 88%-92%) reduced mortality at 90 days, when compared with usual oxygen therapy, in adults receiving mechanical ventilation and supplemental oxygen in the ICU.

Methods

Trial Design and Oversight

UK-ROX was a multicenter, pragmatic, registry-embedded, randomized clinical trial (RCT). The protocol was approved by the South Central-Oxford C Research Ethics Committee (Reference: 20/SC/0423) and the UK Health Research Authority and has been published previously.¹² The UK National Institute for Health and Care Research (NIHR) funded the trial. The Intensive Care National Audit and Research Centre (ICNARC) Clinical Trials Unit managed the trial, with independent oversight by a trial steering committee and a data monitoring and ethics committee. The trial is reported in accordance with the 2020 Consolidated Standards of Reporting Trials (CONSORT) reporting guidelines for parallel-group randomized trials.¹³

Consent Procedures

In accordance with the approved emergency waiver of consent under the relevant Mental Capacity Acts in England, Wales,

Key Points

Question Does reducing exposure to supplemental oxygen by using a reduced peripheral oxygen saturation (SpO₂) target of 90% (range, 88%-92%) decrease 90-day all-cause mortality in mechanically ventilated adult patients receiving supplemental oxygen in intensive care units?

Findings This randomized clinical trial of 16 500 participants found no statistically significant difference between the groups with 35.4% of patients randomized to conservative oxygen therapy having died by 90 days compared with 34.9% of patients receiving usual oxygen therapy.

Meaning The findings do not support an approach of reducing oxygen exposure by targeting an SpO₂ of 90% in mechanically ventilated adults receiving oxygen in an intensive care unit.

and Northern Ireland, a “research without prior consent” approach was used, which allowed agreement to be obtained from a personal or nominated consultee as soon as appropriate following randomization. If the patient regained mental capacity, informed consent was obtained. If the patient had capacity prior to randomization, verbal consent could be obtained. All data collected up to refusal or withdrawal of consent were retained. In addition, approvals were obtained to allow the primary outcome to be collected on all participants, other than those who requested that all data be removed. Secondary outcomes were available for those who provided consent to allow data linkage. All procedures are in the eMethods section of [Supplement 2](#).

Trial Sites and Study Population

The trial was conducted in 97 National Health Service adult, general ICUs that participate in the Case Mix Programme (CMP) national clinical audit for adult ICUs in England, Wales, and Northern Ireland. The study population comprised critically ill adults aged 18 years or older, enrolled within 12 hours of meeting the following criteria: receiving invasive mechanical ventilation following an unplanned admission to the ICU or where invasive mechanical ventilation was started in the ICU and receiving supplemental oxygen. Patients were excluded if randomized to UK-ROX in the previous 90 days, if in receipt of extracorporeal membrane oxygenation, or if the treating clinician considered that the intervention was either clinically indicated or contraindicated.

Randomization and Blinding

Randomization occurred as soon as possible after confirmation of eligibility. Participants were allocated 1:1, via a concealed central 24-hour telephone or web randomization system, to conservative oxygen therapy or usual oxygen therapy. Randomization used permuted blocks with variable block sizes, stratified by site and the following (hierarchical) diagnostic subgroups: hypoxic-ischemic encephalopathy (HIE); sepsis; acute brain injury (except HIE); or none of the prespecified subgroups, as defined by the treating clinician. Treatment allocation was not blinded.

Interventions

Aiming to minimize exposure to supplemental oxygen, participants in the conservative oxygen therapy group received the lowest fraction of inspired oxygen (FIO_2) possible to maintain their pulse oximeter derived SpO_2 at 90%. Sites were instructed to set monitor alarms to sound if levels fell below 88% and or exceeded 92%, once the patient was within range. The upper limit alarm could be deactivated once the patient was receiving an FIO_2 of 0.21. Deviations were allowed if there were major discrepancies with the arterial blood gas-derived oxygen saturation (SaO_2) and the SpO_2 ; a high FIO_2 was needed to prevent an acute life-threatening illness; or a change in clinical circumstances occurred that would have precluded eligibility to the trial. Adherence to the conservative oxygen therapy group was defined as a reduction in supplemental oxygen when the SpO_2 was higher than 92% or an increase in oxygen when lower than 88%. Clinicians were permitted to alter other therapies as required. Full descriptions of the assessment of treatment exposure and adherence are in the eMethods section of [Supplement 2](#).

In the usual oxygen therapy group, participants received supplemental oxygen at the discretion of the treating clinician. No minimum FIO_2 was mandated and no upper SpO_2 limit monitor alarm was set. Interventions were continued until 90 days after randomization or discharge from the ICU, whichever was sooner. If readmitted to the ICU during this period, units were advised to recommence.

Outcome Measures

The primary outcome was all-cause mortality at 90 days after randomization. Secondary outcomes were duration of the ICU and acute hospital stay (censored at 90 days); days alive and free from organ support (respiratory, cardiovascular, or kidney support) at 30 days; mortality at ICU and hospital discharge (censored at 90 days); and 60-day and 1-year mortality. All definitions are in the eMethods section of [Supplement 2](#). The integrated health-economic evaluation will be reported separately.

Sample Size Calculations

Based on CMP, data were collected from April 28, 2017 ($n = 96\,028$) to March 2019, and the Risk II study dataset ranges from April 2014 to March 2016 ($n = 82\,075$).¹⁴ 90-day all-cause mortality was anticipated to be 37% for usual oxygen therapy. Assuming 6% loss to follow-up, a sample size of 16 500 provided 90% power at $P < .05$ to detect an absolute risk reduction of 2.5 percentage points to 34.5% with conservative oxygen therapy. Two interim analyses were performed after 4500 and 10 000 participants using a Haybittle-Peto stopping rule ($P < .001$) for effectiveness or harm.

Data Collection

For efficiency of trial delivery, the majority of data were obtained from linked, routine data sources: the CMP (for baseline data, ICU, and hospital outcomes) and Civil Registrations of Death (for mortality after hospital discharge).¹⁵ To understand oxygen administration and adherence to the intervention, SpO_2 and FIO_2 were collected hourly on a

sample of participants selected for enhanced data collection for 10 days after randomization. Total exposure to supplemental oxygen was calculated by the amount administered above room air (FIO_2 of 0.21). For example, 1 hour with an FIO_2 of 1.0 or 2 hours with an FIO_2 of 0.605 are calculated as one 100%-equivalent hour because both equate to an additional 79% of oxygen (the maximum for a single hour above room air). Enhanced data were collected for approximately 15% of participants: the first 10 participants at each site, to ensure the protocol was being adhered to, followed by a random sample of 10% of subsequent participants. If adherence was deemed unacceptable, enhanced data collection was extended.

Statistical Analysis

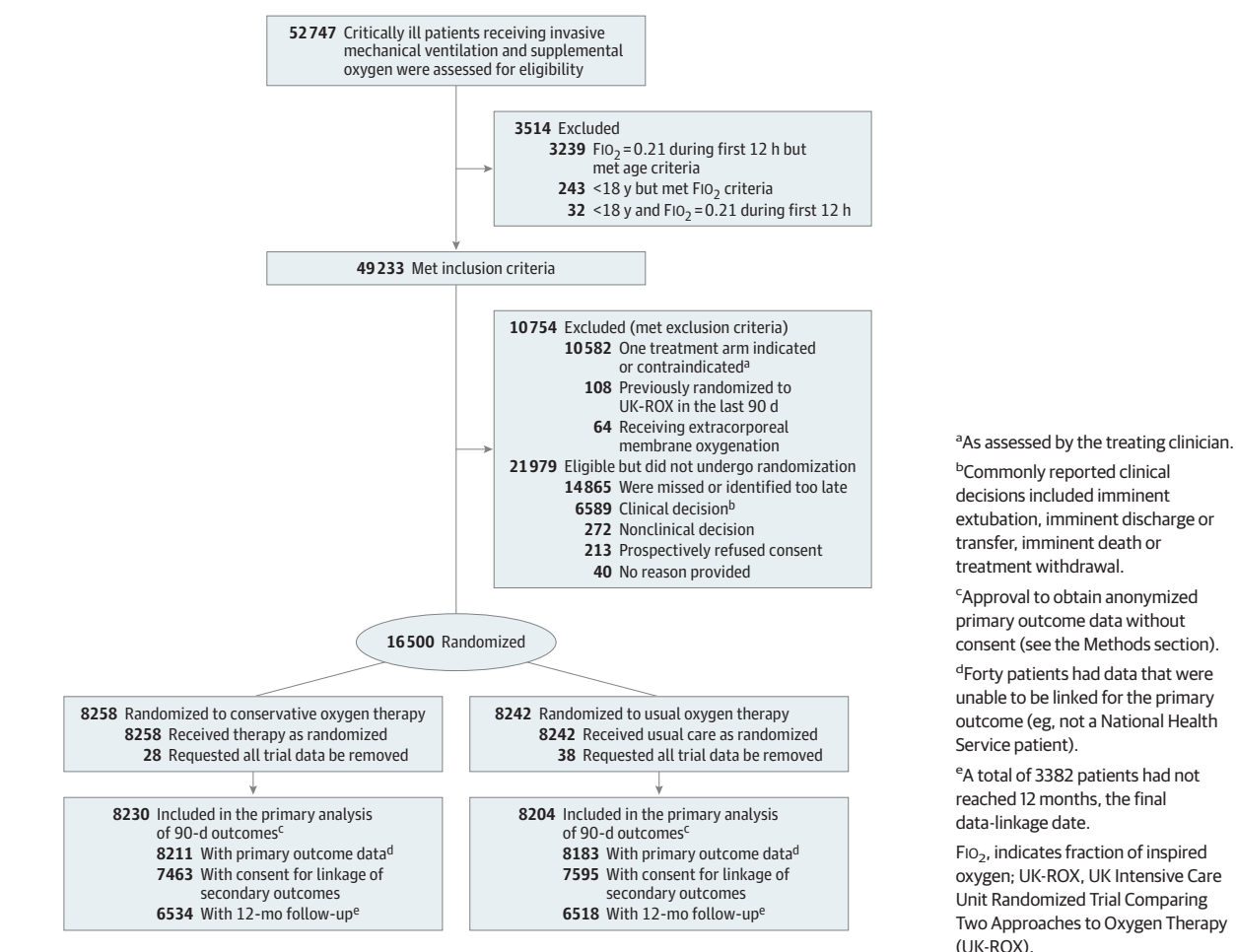
Participants were analyzed according to their randomized group, following a prespecified statistical analysis plan ([Supplement 1](#)). All statistical tests were 2-sided with significance set at $P < .05$ unless otherwise specified. Effect estimates are reported with 95% CIs. There was no adjustment for multiple testing.

The primary analyses were adjusted for the stratification variables (site and diagnostic subgroup) and for additional prespecified baseline covariates that were deemed strong predictors of outcome (age, SpO_2 at randomization, $\text{PaO}_2\text{:FIO}_2$ ratio at randomization, confirmed or highly suspected COVID-19, and date of randomization). Effects were estimated using logistic regression for binary outcomes, Fine-Gray subdistribution hazards regression for durations of ICU and hospital stay among survivors (with death as a competing risk), and ordered logistic regression for days alive and free of organ support. Time to death was analyzed using Cox proportional hazards regression with censoring at the earliest of withdrawal, 365 days or the end of the trial. All models accounted for clustering by site and were adjusted for diagnostic subgroup (stratification variable) and for the same prespecified baseline predictors of outcome. Risk differences and relative risks were estimated using marginal standardization.¹⁶ Multivariable imputation by chained equations was used to account for missing data, incorporating at least the primary outcome as an auxiliary variable to support imputation of secondary outcomes.

The primary outcome was also analyzed by prespecified subgroups (diagnostic subgroup, confirmed or highly suspected COVID-19 vs not, and ethnic group). For each subgroup, the primary outcome analysis was repeated including an interaction between conservative oxygen therapy and the subgroup variable (for multinomial subgroup variables, 1 interaction term for each dummy variable). Subgroup effects were tested (jointly for any subgroup variables with >2 categories) on the odds ratio (OR) scale.

Additional post hoc analyses included subgroup analyses by severity of illness (tertile of predicted risk of death; tertile of Acute Physiology and Chronic Health Evaluation [APACHE] II score¹⁷; and categories of $\text{PaO}_2\text{:FIO}_2$ ratio aligned with acute respiratory distress syndrome definitions)¹⁸ and by data collection subset (the first 10 patients and random enhanced data collection sample vs standard data collection).

Figure 1. Screening, Randomization, and Follow-Up of Participants in the UK-ROX Trial



All statistical analyses were conducted in Stata/MP version 18.0 (StataCorp).

Results

Sites and Participants

A total of 52 747 critically ill patients receiving invasive mechanical ventilation were screened at the 97 sites from May 4, 2021, through November 27, 2024, of whom 38 479 were potentially eligible and 16 500 were enrolled (Figure 1 and eFigure 1 and eTable 1 in Supplement 2). Sixty-six participants (0.4%) requested removal of all data and were excluded from the analysis. The primary outcome was unable to be determined for another 40 participants (0.2%) because data could not be linked. These patients were included in the multiply-imputed primary analysis of 16 434 participants (8230 in the conservative oxygen therapy, 8204 in the usual oxygen therapy; 8211 and 8183 with primary outcome recorded, respectively). Ninety-day follow-up was completed in February 2025, with linkage to death registrations conducted in March 2025, at which time 13 052 participants had reached 12 months' follow-up.

The randomized groups were similar at baseline (Table 1; eTable 2 in Supplement 2) and were representative of the wider ICU population on key demographic factors (eTable 3 in Supplement 2). In both groups, the median age was 60 (IQR, 48-71) years and 5652 (38.2%) were female. Participants were randomized shortly after first receiving invasive mechanical ventilation in the ICU, with a median time to randomization of 5 (IQR, 2-8) hours in both groups. Prior to randomization, median SpO_2 was 97% (IQR, 94%-99%) in the conservative oxygen therapy group and 96% (IQR, 94%-99%) in the usual oxygen therapy group. A total of 1504 participants (9.2%) were admitted due to HIE, 5443 (33.1%) due to sepsis, and 363 (2.2%) due to acute brain injury, with the remaining 9124 (55.5%) not in any of the prespecified subgroups of interest. Of all participants, 1099 (6.7%) had confirmed or highly suspected COVID-19 on enrollment.

Of the 16 434 participants included in the primary analysis, 2489 (1252 conservative oxygen therapy, 1237 usual oxygen therapy) were selected for enhanced data collection. Due to the timing of the start of the trial, there was a higher proportion of patients who had confirmed or highly suspected COVID-19 in the nonrandom first 10 patients in the enhanced

Table 1. Patient Baseline Characteristics by Oxygen Therapy Group

	Oxygen therapy group, No./ total (%)	
	Conservative (n = 8230)	Usual (n = 8204)
Age, median (IQR), y [No.]	60 (48-71) [8230]	60 (48-71) [8204]
Sex ^a		
Female	2803/7340 (38.2)	2849/7465 (38.2)
Male	4537/7340 (61.8)	4616/7465 (61.8)
Ethnic group ^a		
Asian	263/7340 (3.6)	243/7465 (3.3)
Black	138/7340 (1.9)	153/7465 (2.0)
White	6072/7340 (82.7)	6207/7465 (83.1)
Mixed	52/7340 (0.7)	60/7465 (0.8)
Other or not stated ^b	815/7340 (11.1)	802/7465 (10.7)
BMI		
<18.5	264/7111 (3.7)	259/7225 (3.6)
18.5-<25	2291/7111 (32.2)	2299/7225 (31.8)
25-<30	2129/7111 (29.9)	2250/7225 (31.1)
30-<40	1918/7111 (27.0)	1881/7225 (26.0)
≥40	509/7111 (7.2)	536/7225 (7.4)
Preexisting severe respiratory disease ^c	171/7310 (2.3)	172/7436 (2.3)
Prior length of hospital stay, median (IQR), d [No.]	1 (1-3) [7293]	1 (1-3) [7419]
Prior duration of invasive mechanical ventilation in ICU, median (IQR), h [No.] ^d	5 (2-8) [8230]	5 (2-8) [8204]
Current or suspected diagnosis ^e		
Sepsis	2738/8230 (33.3)	2705/8204 (33.0)
HIE	754/8230 (9.2)	750/8204 (9.1)
Acute brain injury (except HIE)	183/8230 (2.2)	180/8204 (2.2)
None of the prespecified subgroups	4555/8230 (55.3)	4569/8204 (55.7)
Confirmed or highly suspected COVID-19	536/8230 (6.5)	563/8204 (6.9)
Spo ₂ , median (IQR), % [No.]	97 (94-99) [8230]	96 (94-99) [8204]
Spo ₂ , %		
<88	138/8230 (1.7)	146/8204 (1.8)
88-92	995/8230 (12.1)	1085/8204 (13.2)
93-95	1979/8230 (24.0)	1913/8204 (23.3)
>95	5118/8230 (62.2)	5060/8204 (61.7)
Fio ₂ , median (IQR) [No.]	0.45 (0.35-0.60) [8230]	0.45 (0.35-0.60) [8204]
Pao ₂ , median (IQR), mm Hg [No.]	90 (75-116) [7638]	89 (74-114) [7620]
Pao ₂ :Fio ₂ ratio		
≤100	933/7638 (12.2)	936/7620 (12.3)
>100 to ≤200	2635/7638 (34.5)	2664/7620 (35.0)
>200 to ≤300	1978/7638 (25.9)	1977/7620 (25.9)
>300	2092/7638 (27.4)	2043/7620 (26.8)
ICNARC _{H-2023} model, mean (SD) [No.] ^f	0.35 (0.29) [6882]	0.34 (0.29) [7014]
APACHE II score, median (IQR) [No.] ^g	16 (12-21) [7317]	16 (12-21) [7437]

Abbreviations: APACHE II, Acute Physiology and Chronic Health Evaluation II; BMI, body mass index, calculated as weight in kilograms divided by height in meters squared; Fio₂, fraction of inspired oxygen; HIE, hypoxic-ischemic encephalopathy; ICNARC, Intensive Care National Audit & Research Centre; ICU, intensive care unit; Pao₂, partial pressure of oxygen in the arterial blood; Spo₂, peripheral oxygen saturation.

SI conversion factor: To convert Pao₂ and Pao₂:Fio₂ ratio to kPa, multiply by 0.133.

^a Data are available when consent is in place for linkage for the following variables: sex; ethnic group; body mass index; preexisting severe respiratory disease; prior length of stay in hospital; ICNARC-2023 model; and APACHE II score.

^b Other includes those in the Chinese ethnic group and those not in groups collected according to the National Health Service Data Dictionary; category terms reflect those used in the study.

^c Shortness of breath with light activity due to pulmonary disease and evident within 6 months before admission.

^d Calculated at admission to critical care.

^e For stratified randomization, hierarchical classification was used to select at most 1 subgroup for each patient (highest to lowest priority: HIE, sepsis, acute brain injury except HIE).

^f ICNARC model, H-2023 predicted death, calculated using physiological measures, age, medical history, dependency, cardiopulmonary resuscitation prior to admission, mechanical ventilation receipt, and source of and primary reason for admission. Other than Pao₂ and Fio₂ values, physiological measures (first 24 hours in critical care).

^g Score range, 0-71; higher scores, greater severity, calculated using physiological measures (first 24 hours in critical care), age, and previous health status. Other than Pao₂ and Fio₂ values.

data collection group from each site (13.4%) compared with the subsequent randomly selected patients (5.3%) and the standard data collection patients (6.4%). Otherwise, these groups were similar (eTable 4 in [Supplement 2](#)).

Oxygen Exposure

Exposure to supplemental oxygen was lower for participants in the conservative oxygen therapy group, with a mean (SD) of the median FIO_2 of 0.31 (0.14) compared with 0.35 (0.15) for those in the usual oxygen therapy group. Total exposure to supplemental oxygen was 29.3% lower in participants in the conservative oxygen therapy group than in those in the usual oxygen therapy group (20.3 vs 28.7, 100%-equivalent hours, respectively; difference, -8.4 hours; 95% CI, -10.8 to -6.0; **Figure 2** and eFigures 2 and 3 and eTable 5 in [Supplement 2](#)).

Arterial oxygenation was lower in the conservative oxygen therapy group with a mean (SD) of the median SpO_2 of 93.3% (2.8%) and the median PaO_2 of 71.5 (13.9) mm Hg compared with 95.1% (2.4%) mm Hg and 79.5 (17.9) mm Hg, respectively, for the usual oxygen therapy group. Participants in the conservative oxygen therapy group spent a mean (SD) of 62.6 (62.3) hours within the SpO_2 target range (88%-92%) compared with 27.2 (39.1) hours for the usual oxygen therapy group. Although higher than the target range, a mean (SD) of 39.7 (55.1) hours was spent receiving no supplementary oxygen in the conservative oxygen therapy group compared with 26.1 (45.1) hours in the usual oxygen therapy group. Participants had an SpO_2 lower than 88% for a mean (SD) of 3.2 (6.5) hours in the conservative oxygen therapy group and 2.3 (7.3) hours in the usual oxygen therapy group (**Figure 2**; eFigures 2 and 3 and eTable 5 in [Supplement 2](#)). Separation was maintained across all enhanced data collection patient groups (eTable 6 in [Supplement 2](#)) and when plotted across calendar time and patient sequence to understand any potential contamination into usual care (eFigure 4 in [Supplement 2](#)).

Adherence to the Protocol

Of participants allocated to conservative oxygen therapy and selected for enhanced data collection, 526 (42.1%) had 1 or more periods of nonadherence, representing 10.6% of their time in the ICU. There were a total of 2271 periods of nonadherence of 3 hours or more. The reasons included staffing issues and lack of awareness ($n = 857$), other clinical priorities ($n = 413$), responding to low PaO_2 ($n = 127$), clinical decision to suspend intervention (not supported by the protocol; $n = 265$), and reason not documented ($n = 609$).

Primary and Secondary Outcomes

In the conservative oxygen therapy group, 2908 participants (35.4%) died compared with 2858 (34.9%) in the usual oxygen therapy group. After adjustment for prespecified baseline variables, the risk difference was 0.7 percentage points (95% CI, -0.7 to 2.0; $P = .28$; **Table 2**) compared with an unadjusted risk difference of 0.5 percentage points (95% CI, -1.0 to 2.0). There were no missing data among the baseline variables used for adjustment, other than for the $\text{PaO}_2\text{:FIO}_2$ ratio

which was singly imputed from the $\text{SpO}_2\text{:FIO}_2$ ratio (eTable 7 in [Supplement 2](#)).

Secondary mortality outcomes at ICU discharge, 60 days, and 1 year were not significantly different by treatment group (**Table 2**). Time to death (adjusted hazard ratio, 1.01; 95% CI, 0.96-1.05; eFigure 4 in [Supplement 2](#)), and duration of ICU and acute hospital stay among survivors were not significantly different between the groups. Survivors in the conservative oxygen therapy group stayed a median of 20 (IQR, 11-40) days in acute hospital compared with 21 (IQR, 10-42) days in the usual oxygen therapy group (hazard ratio, 0.98; 95% CI, 0.94-1.02). Days alive and free from organ support at 30 days were not significantly different in the conservative oxygen therapy group compared with the usual oxygen therapy group (proportional OR, 1.01; 95% CI, 0.96-1.07; **Table 2** and eFigure 6 and eTable 8 in [Supplement 2](#)). In the conservative oxygen therapy group, 58 participants (0.7%) had serious adverse events reported compared with 29 (0.4%) in the usual oxygen therapy group (eTable 9 in [Supplement 2](#)).

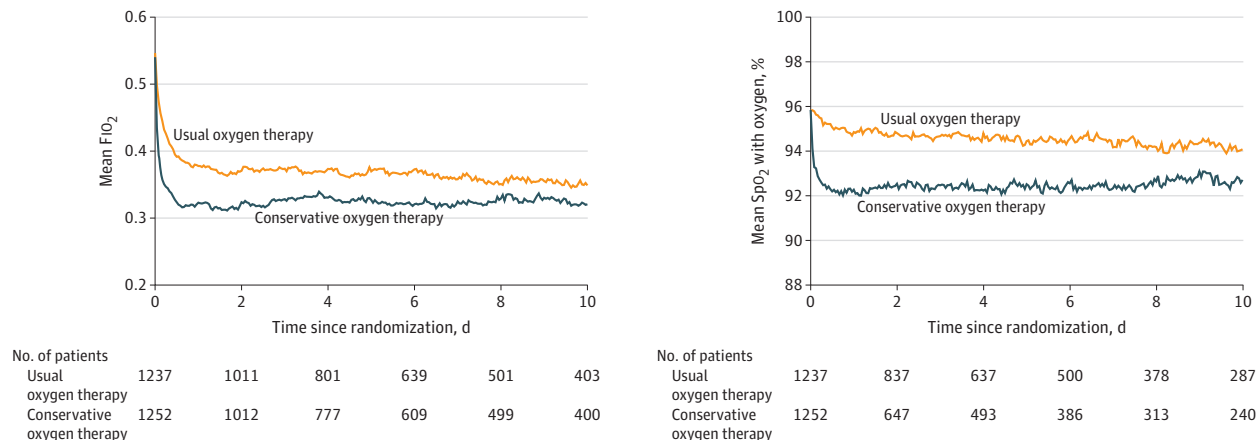
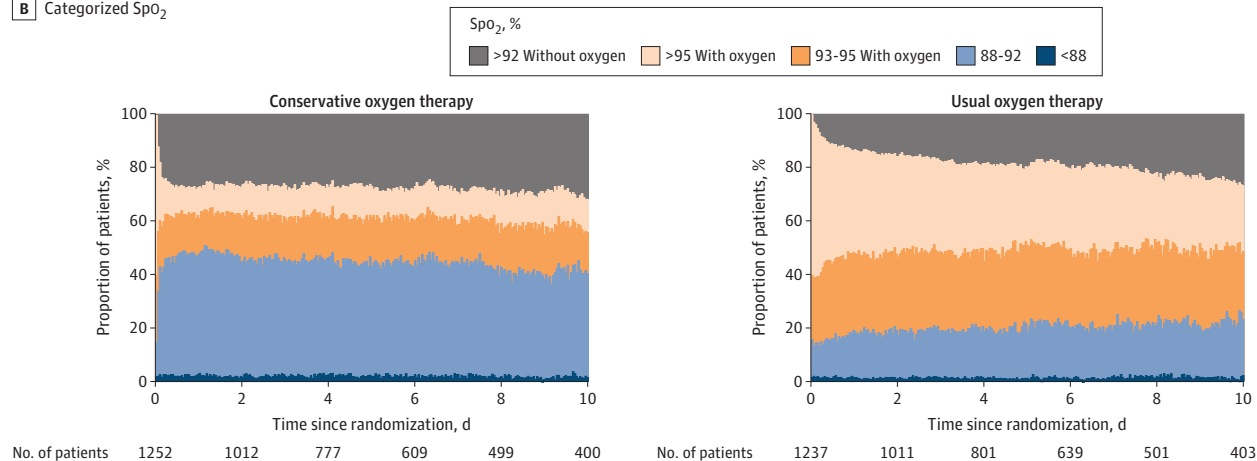
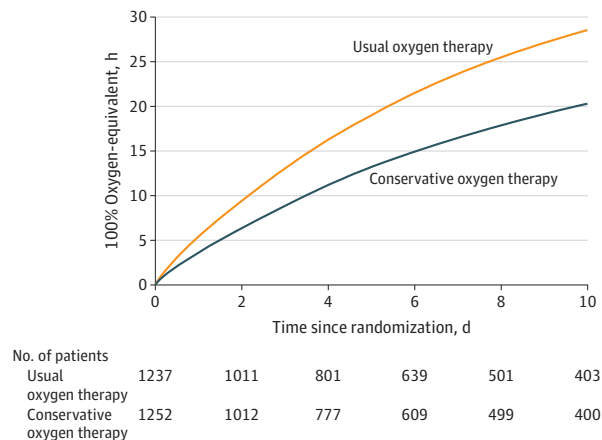
Tests for interaction were not statistically significant for diagnostic subgroup, confirmed or highly suspected COVID-19, or ethnic group (**Figure 3**), or for post hoc subgroups by severity of illness (eFigure 7 in [Supplement 2](#)). In the post hoc analysis by data collection subgroup, there was weak evidence of increased harm from conservative oxygen therapy among the first 10 patients in each site but no difference for the random enhanced data collection sample compared with standard data collection (eFigure 7 in [Supplement 2](#)).

Discussion

In this RCT of mechanically ventilated critically ill adult patients receiving supplementary oxygen in UK ICUs, minimizing oxygen exposure through conservative oxygen therapy did not reduce all-cause mortality at 90 days compared with usual oxygen therapy. The observed reduction in exposure to oxygen translated to an adjusted absolute increase of 0.7 percentage points in mortality at 90 days with a 95% CI from a 0.7-percentage point reduction to a 2.0-percentage point increase. We found no significant differences in prespecified or exploratory subgroup analyses of the primary outcome or in any of the secondary outcomes.

These findings are consistent with other RCTs of conservative oxygen therapy reporting mortality⁴⁻⁸ and a recent systematic review and meta-analysis of trials.¹¹ The trial design differed from others by having a usual care comparator rather than protocolized liberal oxygen therapy. Prior data had shown that the average SpO_2 of adult ICU patients in the UK was approximately 96%¹⁹ and the findings of our feasibility RCT indicated that clinicians would be unwilling to maintain a minimum FIO_2 in the comparator group.²⁰ Arterial oxygenation data from the usual oxygen therapy group suggest a more conservative approach to oxygen administration than was observed in other countries in the last 2 decades.^{2,21} Our findings add to the understanding of oxygenation targets for critically ill patients by evaluating oxygen therapy in more participants than all prior trials combined. The results of an ongoing larger RCT

Figure 2. Oxygen Exposure and Arterial Oxygenation by Treatment Group in Patients With Enhanced Data Collection

A Separation in FiO_2 and SpO_2 when receiving supplemental oxygen**B** Categorized SpO_2 **C** Cumulative exposure to supplemental oxygen over first 10 d after randomization

FiO_2 indicates fraction of inspired oxygen; SpO_2 , peripheral oxygen saturation. Patients whose SpO_2 ranged from <88% to 92% were included regardless of whether they were receiving supplemental oxygen or not.

comparing conservative oxygen therapy to protocolized liberal oxygen therapy (minimum acceptable FiO_2 of 0.3) are awaited.²²

The goal of conservative oxygen targets is to minimize exposure to oxygen, yet most trials have not achieved their specified arterial oxygenation targets.²³ In part, this is because SpO_2

Table 2. Primary and Secondary Clinical Outcomes

Outcome	Conservative oxygen therapy group	Usual oxygen therapy group	Adjusted effect (95% CI)		P value
			Estimate available case ^a	Multiply imputed ^b	
Primary outcome					
90-d Mortality, No./total (%)	2908/8211 (35.4)	2858/8183 (34.9)	RD, 0.7 (−0.6 to 2.1)	RD, 0.7 (−0.7 to 2.0)	.28
			RR, 1.02 (0.98 to 1.06)	RR, 1.02 (0.98 to 1.06)	
			OR, 1.04 (0.97 to 1.11)	OR, 1.04 (0.97 to 1.11)	
Secondary outcomes					
Duration of ICU stay, median (IQR), d [No.]					
Overall	6.6 (3.1 to 13.3) [7333]	6.8 (3.1 to 13.8) [7448]			
Survivors	7.3 (3.6 to 14.9) [5211]	7.7 (3.8 to 15.3) [5290]	sHR, 1.00 (0.96 to 1.04)	sHR, 1.00 (0.96 to 1.04)	.97
Nonsurvivors	4.9 (1.7 to 10.4) [2122]	4.6 (1.7 to 9.8) [2158]			
Duration of acute hospital stay, median (IQR), d [No.]					
Overall	14 (7 to 30) [7323]	14 (7 to 31) [7434]			
Survivors	20 (11 to 40) [4791]	21 (10 to 42) [4906]	sHR, 0.98 (0.94 to 1.02)	sHR, 0.98 (0.94 to 1.02)	.27
Nonsurvivors	7 (3 to 14) [2532]	7 (3 to 13) [2528]			
Days alive and free from organ support at 30 d, median (IQR), d ^c	16 (−1 to 25) [7327]	16 (−1 to 25) [7444]	POR, 1.00 (0.95 to 1.06)	POR, 1.01 (0.96 to 1.07)	.64
30-d Mortality, No./total (%)	2435/7449 (32.7)	2427/7573 (32.0)			
Survivors free from organ support, d [No.]	23 (16 to 26) [4933]	23 (15 to 26) [5054]			
Mortality at ICU discharge, No./total (%)	2122/7334 (28.9)	2161/7453 (29.0)	RD, 0.2 (−1.2 to 1.6)	RD, −0.1 (−1.3 to 1.1)	.94
Mortality at acute hospital discharge, No./total (%)	2533/7335 (34.5)	2535/7458 (34.0)	RD, 0.9 (−0.6 to 2.3)	RD, 0.5 (−0.8 to 1.9)	.46
60-d Mortality, No./total (%)	2637/7449 (35.4)	2617/7573 (34.6)	RD, 1.1 (−0.2 to 2.5)	RD, 0.8 (−0.6 to 2.2)	.25
1-y Mortality, No./total (%)	2295/5636 (40.7)	2314/5755 (40.2)	RD, 1.0 (−0.7 to 2.6)	RD, 3.3 (−0.7 to 7.3)	.34

Abbreviations: ICU, intensive care unit; OR, odds ratio; POR, proportional odds ratio; RD, risk difference; RR, risk ratio; sHR, subdistribution hazard ratio, calculated using the Fine and Gray method to account for competing risk of death.

^a Regression models were adjusted for stratified randomization factors of ICU site and diagnostic stratum, plus confirmed or highly suspected COVID-19 and restricted cubic splines of age, peripheral oxygen saturation (SpO₂), partial pressure of arterial oxygen to fraction of inspired oxygen (PaO₂:FiO₂) ratio, and date of randomization.

^b Primary analysis was adjusted as above, with multiple imputation of missing

data, except for duration of ICU and acute hospital stay, or which patients with missing time to discharge were included as censored 1 hour after randomization.

^c Ordinal composite outcome with patients who died on or before day 30 assigned the worst possible score of −1. Surviving patients were ranked according to the number of calendar days on which any respiratory, cardiovascular, or renal support was received at any time during that day, starting from day 1 (the day of randomization) up to and including day 30. The following rows show individual components: 30-day mortality and days free from organ support at 30 days among survivors.

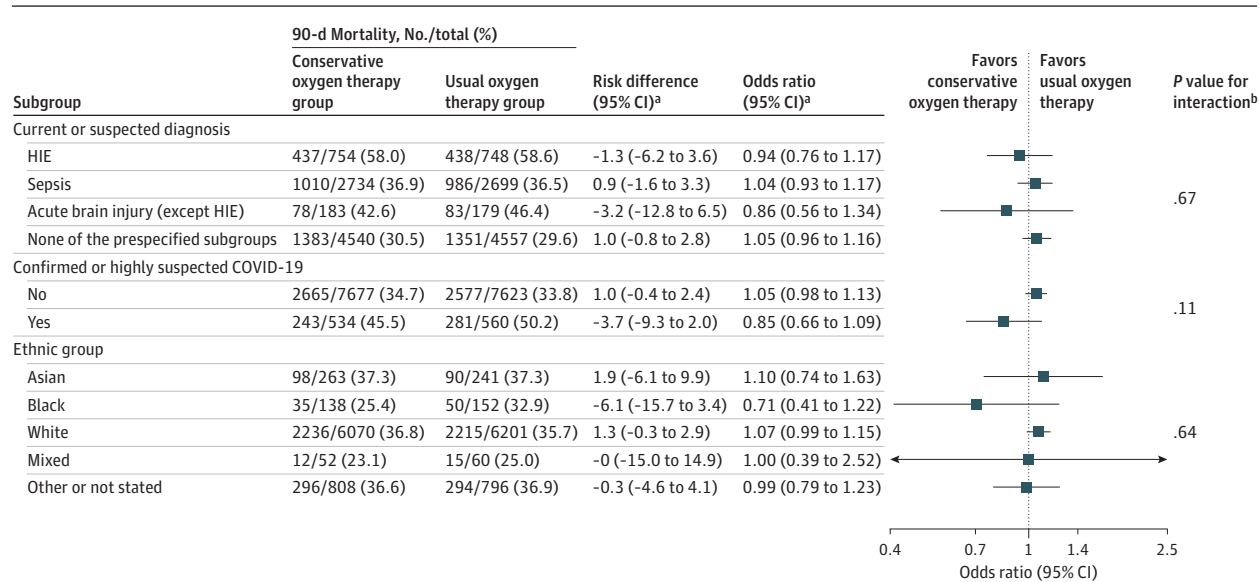
often exceeds the upper target limit, even when no additional oxygen is given. Separation was observed between groups in all oxygen metrics; however, it was smaller than reported in other similar trials (eTable 10 in Supplement 2). This was in part due to usual care being more conservative than liberal comparator groups in previous trials, potentially reflecting the more recent trend toward giving less oxygen in the ICU.

It is plausible that oxygen therapy has a differing effect according to patients' characteristics.²⁴ Machine learning techniques have demonstrated that patients predicted to benefit from lower oxygenation targets had a higher prevalence of acute brain injury, whereas patients predicted to benefit from higher targets had a higher prevalence of sepsis.²⁵ In a trial enrolling patients with COVID-19, a lower oxygenation target was beneficial.⁹ The direction of the signals detected in our

COVID-19 and sepsis cohorts align with these. An analysis of heterogeneity of treatment effect in the UK-ROX trial using machine learning techniques was prespecified in the statistical analysis plan and will be reported separately.

The trial has several strengths. The size of the trial ensured adequate power to detect the small absolute risk reduction hypothesized to be associated with conservative oxygen therapy. Additionally, the high precision of the results means that a clinically important reduction of mortality from conservative oxygen therapy is very unlikely. The trial benefited from an efficient design, using linkage to available registries to significantly reduce its cost, thereby supporting a large sample. This sample was highly representative of the whole potentially eligible UK ICU population enhancing generalizability. Patients were rapidly enrolled into the trial following

Figure 3. Subgroup Analyses of Primary Outcome



^aAdjusted for site, diagnostic subgroup, age, peripheral oxygen saturation (SpO₂), partial pressure of arterial oxygen to fraction of inspired oxygen (PaO₂:FIO₂) ratio, confirmed or highly suspected COVID-19, and date of randomization. A negative risk difference indicates an observed lower mortality in the conservative oxygen therapy groups.

^bP value are from tests of interactions in the odds ratio in the adjusted multilevel logistic regression model.

HIE indicates hypoxic-ischemic encephalopathy.

eligibility, reducing the likelihood of inappropriate oxygen therapy while in the ICU prior to randomization.

Limitations

Like other trials of conservative oxygen therapy, clinicians and patients could not be blinded to treatment intervention, and other aspects of care were at the discretion of clinicians. However, the primary outcome of the trial is unlikely to be subjected to bias. Clinicians excluded a sizable proportion of potential participants due to the intervention being either indicated or contraindicated, which could have meant patients who may have benefited were not included. It also may reduce the generalizability of the findings. A proportionate approach to data collection was necessary to allow a trial of this size to be delivered; however, it means that oxygenation cannot be confirmed for all patients in the trial. The use of a usual care comparator, essential because clinicians were unwilling to give potentially unnecessary oxygen therapy,

could increase the risk of contamination; however, we found no evidence of this because separation was sustained over the trial. Other than adjusting for the baseline SpO₂ and the PaO₂:FIO₂ ratio, prior oxygen administration could not be accounted for and, for some patients, could have been significant. Regarding protocol adherence, among enhanced data patients, episodes of nonadherence occurred in 42.1%. However, these episodes accounted for a small proportion of ICU hours and overall separation on oxygen exposure between groups was substantial.

Conclusion

In mechanically ventilated adults admitted to an ICU, minimizing oxygen exposure by targeting an SpO₂ of 90% did not reduce all-cause mortality at 90 days compared with usual oxygen therapy.

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The UK-ROX Investigators are listed in [Supplement 3](#).

Data Sharing Statement: See [Supplement 4](#).

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